MBB — park homophily, replay draft selection on real rosters

Last synced: 2026-08-28

Audience: Charles (slow read / print). Summary of Aug 28 conversation with Alex after PD28 porch + calibration work.

Standalone: On men's college basketball (MBB), we treat homophily as ~0, keep real players on real teams, fit score knobs on that frozen world, simulate who gets drafted, and compare the resulting HERO plot to empirical data.

In plain language first

We spent a lot of time on **homophily** — the idea that similar players end up on the same teams. On MBB, the data keep saying: **there isn't much of that.** Empirical ρ is about **zero**. So does **found assortativity** (H_{sort}): only about **6%** of ability variance is explained by which team you're on. Alex's reaction: *maybe homophily needs a rethink someday* — but **not this week.**

The productive move for MBB is narrower and honest:

1. **Take the real world as given** — real players, real teams, real season abilities ($\hat{A}_i$ from PPM, etc.).
2. **Do not** build a fake league and try to re-assign people with ρ.
3. **Fit** how drafting *scores* people: γ^*, λ^*, and temperature (how sharp the ranking is).
4. **Simulate** who gets drafted using those fitted knobs.
5. **Plot** the same HERO readout we already use empirically — draft rate vs leave-one-out (LOO) teammate quality — and ask: **does the model shape look like the data?**

That is **score → select** on a **fixed roster**. It is not "we rebuilt college basketball from scratch." It is: *given this locker room, does our draft mechanism produce the right kind of crowding story?*

Baby examples

Example A — The restaurant brigade (fixed ASSIGN)

Imagine a kitchen with **fixed stations** (teams). Everyone's already placed. You are **not** asking "should the sauté cook swap stations?" (ρ, homophily). You're asking: "Given who is already on the line, **who gets promoted to head chef**?" (draft / advancement).

- **ASSIGN** = who works which station → **frozen** (empirical rosters).
- **SCORE** = how valuable each cook looks given their peers → fit λ, γ, t.
- **SELECT** = who actually gets promoted → simulate draftees.
- **HERO** = graph: "As local talent around you goes up, does promotion rate go up forever, peak and fall, or something else?"

On MBB with PPM, the empirical HERO is often **flat or monotone** — not a dramatic inverted-U. The **generative** model *can* show an inverted-U at some (λ, t). The replay test is whether **calibrated** sim draftees land in the same **shape family** as data.

Example B — Why $\rho \approx 0$ is a parking ticket, not a failure

Suppose team assignment were random. Then $H_{\text{sort}} \approx 0$. Our bracket says $\rho^* \approx 0$. That is **consistent**: "don't simulate strong homophily on this domain."

Trying to **force** ρ to move when H_{sort} is flat is like tuning a volume knob that isn't wired to the speaker. You can turn it; nothing useful happens. **Park it.**

Example C — What "generate draftees" means here

Not: invent 5,000 new players and assign them to Duke and Kansas.

Yes: for each **real** 2009–2021 player-season on a **real** roster, compute a draft probability or run a selection rule, draw **simulated** draft flags Y_i^{sim}, then bin by LOO pool quality exactly like Pass A.

Compare:

- Empirical: $\hat{P}(Y{=}1 \mid \text{LOO bin})$
- Simulated: $\hat{P}(Y^{\text{sim}}{=}1 \mid \text{LOO bin})$

Same bins. Same aperture. Side by side.

What Alex proposed (one paragraph)

Park homophily for MBB. Instead of generating a league from an A_i distribution and using ρ for assignment, **use empirical data and empirical teams**, fit λ^*, γ^*, and temperature *, plug those into the model, **generate draftees**, and inspect the **HERO plot**. Does that make sense? **Yes** — and it aligns with what PD21 MLE already started (fit on fixed rosters). The missing piece is a clean **generative SELECT → HERO** replay, not another ρ bracket.

Why this is the right move for MBB

Old wind-tunnel habit	Problem on MBB
Draw synthetic A_i, LG-assign with ρ, match H_{sort}	$\rho^* \approx 0$; bracket has **no slope**
Full synthetic league	ASSIGN layer adds noise, not identifiable signal

Alex proposal	What it buys
Empirical rosters + empirical $\hat{A}_i$	No fake homophily; no ρ algebra
Fit λ^*, γ^*, t^* on that panel	SCORE layer still has bite
Generate draftees → HERO	Tests **SELECT** vs empirical Pass A

Army can keep the full ASSIGN → SCORE → SELECT ladder later. **MBB** becomes: *empirical environment, generative selection.*

What we already have (not starting from zero)

Reigning hero sandbox, 2009–2021, min20, mg10, PPM z:

Piece	Status	Where
$\rho^* \approx 0$ (H_sort bracket)	Done	`population_sandbox/reigning_hero/calibration/rho/`
$\gamma^* \approx 19.57$, $\lambda^* \approx 1.30$, $t^* \approx 1.07$	Done	`.../calibration/mle/REIGNING_PD21_draft_bernoulli_mle_2009_2021_mg10_min20_09_21.json`
Gibbs SELECT temperature sweep	Done	`.../calibration/temperature/`
Empirical HERO (last-ps, LOO, EW16)	Done	porch + reigning hero lock in `reigning_hero/README.md`
Sim HERO replay (Gibbs v1)	**Done (Aug 28)**	`reigning_hero/sim_hero/` · `reigning_hero_sim_hero.py`
Gap	**Sensitivity + axis alignment**	LOO in SCORE? multi-seed β_2 bands? F-HERO dual readout

PD28 notes: `transcripts/PD28_notes.md`
Campaign compare: `population_sandbox/reigning_hero/calibration/CAMPAIGN_COMPARE.md`

Binding rules (still apply)

From project binding docs — short version:

1. **Environment** ($L_{\text{net}} = B - D$) $\neq$ **advancement**.
2. **Score** ($S_i = A_i - \lambda L_C$, etc.) $\neq$ **select** (who actually gets drafted).
3. **Hero** = outcomes; do not merge into "one model" in prose.

Alex's plan respects this: fit **score** knobs on fixed rosters; **then** run **select**; **then** read **hero** (HERO plot). Three steps, named separately.

Deep dive — softmax, Bernoulli, Gibbs, and the two "temperatures"

Charles, read this slowly. This is the piece that confused the "open question" wording. Short answer: **you already decided a lot.** Bernoulli and softmax are **not** two competing draft models. Gibbs is **not** a replacement for Bernoulli. They live at **different steps** in the pipeline. You use **both**, for **different jobs**.

The pipeline in one breath (frozen MBB rosters)

Think of four steps on **real** teams:

```
ASSIGN (frozen)  →  SCORE  →  turn scores into chances  →  SELECT  →  HERO plot
empirical teams     S_i, λ, γ      softmax + maybe t*           who drafts?     draft rate vs LOO
```

- **ASSIGN:** Parked for MBB ($\rho \approx 0$). Rosters are **given**.
- **SCORE:** How good is each player **relative to local context**? $S_i = A_i/t - \lambda L^C$ (PD21 board form).
- **Softmax:** Turn scores into **probabilities** p_i that sum to 1 within each draft season.
- **Bernoulli (fitting only):** How we **learn** λ, γ, t by comparing p_i to actual draft outcomes Y_i.
- **SELECT (generative replay):** How we **simulate** new draftees for the HERO plot — top-K, Bernoulli draws, or **Gibbs** K draws.
- **Gibbs** t**:** Temperature on the **SELECT** step only — how noisy the draft lottery is **after** scores are fixed.

Formula sheet (copy-paste reference)

Indices: player i, draft season s, team $j(i)$. Draft indicator $Y_i \in \{0, 1\}$.

Step 0 — Frozen ASSIGN (MBB reigning)

Teams and teammates are **observed**. No grandchild (LG) sim. Homophily $\rho \approx 0$ parked.

Step 1 — Ability and local context

Symbol	Meaning (reigning lock)
A_i	PPM z-score (winsorized LOO panel)
L_i^C	Local crowding / pool context (**MLE uses** `pool_c_smooth_team`; **HERO x-axis uses** `poolq_loo`)
γ	Viability sharpness — enters **construction** of L^C, not inside the softmax line

Step 2 — SCORE (Alex board, PD21 fitting)

PD21 **board logits** (per player-season):

$$\eta_i \;=\; \frac{A_i}{t_{\mathrm{MLE}}} \;-\; \lambda\, L_i^C$$

Important: this is **not** $(A_i - \lambda L_i^C)/t_{\mathrm{MLE}}$. Ability is scaled by t **before** subtracting congestion. (Grandchild SELECT sweeps use the other order — see below.)

Conceptual Alex score (binding doc):

$$S_i \;=\; A_i \;-\; \lambda\, L_i^C$$

Think of η_i as "S_i with an extra $1/t_{\mathrm{MLE}}$ on talent only."

Step 3 — SOFTMAX (within each draft season s)

For all players i in season s:

$$p_i \;=\; \frac{\exp(\eta_i)}{\sum_{k\in s}\exp(\eta_k)} \qquad \Rightarrow \qquad \sum_{i\in s} p_i = 1$$

So p_i are **relative draft weights** in that season, not independent Bernoulli probabilities yet.

Step 4 — BERNOULLI likelihood (how we fit $\lambda, \gamma, t_{\mathrm{MLE}}$)

Treat each Y_i as an independent Bernoulli trial with "success" probability p_i:

$$\ell(\lambda,\gamma,t_{\mathrm{MLE}}) \;=\; \sum_i \Big[Y_i \log p_i + (1-Y_i)\log(1-p_i)\Big]$$

We maximize ℓ over $(\lambda, \gamma, t_{\mathrm{MLE}})$. **Reigning fit:**

$$\gamma^* \approx 19.57, \quad \lambda^* \approx 1.30, \quad t^*_{\mathrm{MLE}} \approx 1.07$$

Softmax + Bernoulli are partners: softmax builds p_i; Bernoulli scores them against real Y_i.

Step 5 — SELECT (generative replay — three options)

After parameters are fixed, simulate draft flags Y_i^{sim}. Let K_s = number of actual draftees in season s.

A. Deterministic top-K

$$Y_i^{\mathrm{sim}} = 1 \quad \text{iff } i \text{ is among the } K_s \text{ largest } p_i \text{ (or } \eta_i) \text{ in season } s$$

B. Bernoulli replay (matches fit likelihood literally)

$$Y_i^{\mathrm{sim}} \sim \mathrm{Bernoulli}(p_i) \quad \text{independently}$$

Class size $\sum_i Y_i^{\mathrm{sim}}$ is **random** (not fixed at K_s).

C. Gibbs rule D (PD20 / temperature sweep — **separate** t)

First compute Alex score (grandchild convention):

$$S_i = A_i - \lambda^* L_i^C$$

Gibbs weights and K_s draws **without replacement**:

$$w_i \propto \exp\!\left(\frac{S_i}{t_{\mathrm{Gibbs}}}\right), \qquad \text{draw } K_s \text{ players with } \Pr(i \text{ picked}) \propto w_i$$

Sweep: $\log_{10} t_{\mathrm{Gibbs}} \in [-3, 3]$ (reigning calibration folder).

The two temperatures — side by side

	t_{MLE} **(PD21)**	t_{Gibbs} **(PD20 SELECT)**
Where	Inside $\eta_i = A_i/t - \lambda L^C$ before softmax	Inside $\exp(S_i/t)$ at SELECT
Fitted?	Yes — $t^* \approx 1.07$	No — swept on a grid
Effect	Sharpens **spread of** p_i from ability	Adds **lottery noise** after scores exist

Step 6 — HERO readout (not part of fit)

Bin players by leave-one-out pool quality $\mathrm{poolq}_i^{\mathrm{LOO}}$ (EW16), plot draft rate vs bin. Compare empirical Y_i to sim Y_i^{sim}. **Score and select do not appear in the HERO formula** — HERO is an outcome diagnostic.

$$\text{HERO bin } b: \quad \hat{r}_b = \frac{\sum_{i \in b} Y_i}{|b|}$$

What are "board logits"? (plain language)

It might sound like jargon, but the idea is simple.

Logits = the **raw scores** you feed into softmax **before** they become probabilities. They can be any real numbers (positive, negative, large, small). Softmax turns them into p_i that sum to 1 within each draft season.

Board logits = Alex's name for those raw scores on the **draft board** — the within-season table of "who gets how much draft mass this year." **"Board"** is whiteboard language from PD21 (`pd21_draft_bernoulli_mle.py`), not a separate math object.

The one formula (same as Step 2)

For player i in draft season s:

$$\eta_i = \frac{A_i}{t_{\text{MLE}}} - \lambda L_i^C$$

We write η_i or logits_i interchangeably. Code: `board_logits(ability, lc, lam=lam, t=t)` returns $A/t - \lambda L^C$.

Piece	Meaning
η_i	Board logit for player i
A_i	Ability (PPM z)
L_i^C	Local crowding (`pool_c_smooth_team` in the MLE)
λ	Congestion weight in the ranking
t_{MLE}	MLE temperature on ability only

Higher η_i → player looks better on the board → **higher** p_i after softmax.

Why "logits" and not just "Alex score"?

Conceptual Alex score is $S_i = A_i - \lambda L_i^C$. Board logits are a **reparameterization** for the draft-probability step:

$$p_i \propto \exp(\eta_i) = \exp(A_i/t_{\text{MLE}}) \cdot \exp(-\lambda L_i^C)$$

So the board **factorizes** into:

- a **talent** term: $\exp(A_i/t_{\text{MLE}})$
- a **congestion penalty**: $\exp(-\lambda L_i^C)$

"Logits" means: the number **inside** the exponential (up to an additive constant per season). Only **differences** between η_i matter for within-season ranking; adding the same constant to everyone does not change p_i.

Then softmax makes proper probabilities (Step 3):

$$p_i = \frac{\exp(\eta_i)}{\sum_{k \in s} \exp(\eta_k)}$$

Bernoulli fitting (Step 4) asks whether real draft outcomes Y_i match these p_i.

Baby numeric example (three players, one season)

Fix $\lambda = 1.3$, $t_{\text{MLE}} = 1.07$:

Player	A_i	L_i^C	$\eta_i = A_i/1.07 - 1.3\, L_i^C$
Star	2.0	0.5	≈ 1.22
Mid	0.5	0.0	≈ 0.47
Role	-0.5	1.0	≈ -1.77

Softmax → roughly $p \approx (0.62,\ 0.33,\ 0.05)$ (sums to 1).

- **Logits** = $(1.22,\ 0.47,\ -1.77)$ — pre-probability board scores
- p_i = $(0.62,\ 0.33,\ 0.05)$ — post-softmax draft shares

What board logits are not

- **Not** probabilities (they do not sum to 1; Role's logit is negative).
- **Not** the same as Gibbs SELECT weights (those use $S_i = A_i - \lambda L^C$ and a **different** t_{Gibbs} at SELECT).
- **Not** $(A_i - \lambda L_i^C)/t_{\text{MLE}}$ — PD21 puts $1/t$ on A **only**, not on the whole score.

One-liner: *Board logits are the draft-board raw scores $\eta_i = A_i/t_{\text{MLE}} - \lambda L_i^C$ that softmax turns into season draft probabilities p_i.*

See also: `_DISPOSABLE_PD25_Alex_board_for_dummies.md` (standalone PD25 glossary).

Baby example — graduate admissions (same logic)

Frozen ASSIGN: Students are already at specific colleges. We are **not** moving them to new schools.

SCORE: Each senior gets a composite S_i from GPA (A_i) minus "how crowded your major cohort is" (λL^C).

Softmax (within application year): Convert scores to **shares** that add to 100%:

- Alice $S = 2.0$ → $p_{\text{Alice}} = 8\%$ of "admit weight"
- Bob $S = 0.5$ → $p_{\text{Bob}} = 2\%$
- … everyone in the year sums to 100%

Those p_i are **not** literal independent admit probabilities in the real world — they are a **mathematical device** so that better scores get more mass in a **closed** draft season.

Bernoulli (how we FIT): PD21 asks: *if each student were independently admitted with probability p_i, how well does that explain who actually got in?* We tune λ, γ, t until the Bernoulli log-likelihood is happy. **This is calibration, not the generative story Alex tells at the whiteboard.**

SELECT (how we GENERATE for HERO replay): When Alex says "generate draftees," he means **pick winners**. Three flavors:

Mechanism	What happens	Draft class size
Top-K	Take the K highest S_i (or p_i)	Exactly K (matches NBA draft count)
Bernoulli replay	Each player drafted with probability p_i independently	**Random** (might be 58 or 62, not always 60)
Gibbs rule D	K draws **without replacement**, weight $\propto \exp(S/t)$	Exactly K, with noise

Gibbs t: If t is **small**, the highest scores almost always win (like top-K). If t is **large**, middling players sometimes slip in — a **noisy lottery** biased toward good scores. PD20 swept $\log_{10} t$ from 10^{-3} to 10^3 to see if **inverted-U HERO shapes** survive stochastic select.

What you already locked (yes, you decided this)

Decision	What it means	Where
Bernoulli + softmax for FITTING	Learn γ^*, λ^*, t^* on fixed empirical rosters	PD21 Alex board · `pd21_draft_bernoulli_mle.py`
Reigning numbers	$\gamma^* \approx 19.57$, $\lambda^* \approx 1.30$, $t^* \approx 1.07$	`reigning_hero/calibration/mle/...json`
Gibbs for SELECT sensitivity	Stochastic draft after score; sweep temperature	PD20 · `grandchild_temperature_select_sweep.py`
Reigning sweep at $\rho = 0$	Inverted-U-like LOO shapes can appear at some $(\lambda, \text{Gibbs } t)$	`reigning_hero/calibration/temperature/`

So when the memo said "Bernoulli **vs** Gibbs," that was **sloppy**. Correct statement:

- **Bernoulli + softmax** = how we **fit** (already done).
- **Gibbs** = one option for how we **select** when generating sim draftees (also already in the codebase for sweeps).

You did **not** fail to choose. You chose **fit with Bernoulli** and **explore select with Gibbs**. The only open piece is: **for the empirical-roster HERO replay v1, which SELECT rule do we wire as default?**

The two temperatures (both called t — cruel but true)

Same letter, **two different knobs**:

Temperature #1 — MLE t^* (in the score / softmax fit)

- **Formula (PD21 board):** $\text{logits}_i = A_i/t - \lambda L^C$, then $p_i = \text{softmax}(\text{logits})$.
- **Role:** Spreads or sharpens **ability** in the ranking **before** converting to p_i.
- **Fitted value (reigning):** $t^* \approx 1.07$.
- **Baby read:** t small → superstar dominates the probability mass; t large → everyone's p_i looks more equal.

This t was **estimated from data** alongside λ and γ. It is **not** the Gibbs sweep grid.

Temperature #2 — Gibbs t (in SELECT only)

- **Formula:** draft weight$_i \propto \exp(S_i/t)$; draw K players without replacement.
- **Role:** How **random** the draft is **after** scores exist.
- **Not** jointly re-fit in PD21 MLE. Chosen by **sweep** (PD20 / reigning temperature folder).
- **Baby read:** $t \to 0$ → always take the top K scores; t huge → almost a random draft among plausible players.

Mnemonic: MLE t = "how much does raw talent separate people in the **probability table**?" **Gibbs t** = "how much **luck** is in who actually gets the K slots?"

Softmax vs Bernoulli — not either/or

They stack:

1. Compute logits from (A_i, L_i^C, λ, t).
2. **Softmax** → p_i (sum to 1 per season).
3. **Bernoulli log-likelihood** → $\sum_i [Y_i \log p_i + (1 - Y_i)\log(1 - p_i)]$.

Script name says it all: `pd21_draft_bernoulli_mle.py` — **Bernoulli** likelihood on **softmax** probabilities.
We **also** report top-K overlap diagnostics (do the highest p_i match actual draftees?) — that's a **check**, not the likelihood.

What is still open (narrowly) for HERO replay v1

Not "softmax vs Bernoulli." Not "did we pick Gibbs ever."
Open: After fixing γ^*, λ^*, MLE t^* on empirical rosters, **which SELECT rule produces Y_i^{sim} for the sim HERO plot?**

Option	Pros	Cons
A. Top-K by p_i	Exact draft class size; simple	Zero randomness; not Gibbs
B. Bernoulli draws from p_i	Matches the **fit** likelihood literally	Variable K; noisy year-to-year
C. Gibbs K-draw with chosen Gibbs t	Matches PD20 generative story; tunable noise	Need one $\log_{10} t$ from sweep (e.g. where inverted-U appeared)

Charles lock (Aug 28): Gibbs K-draw (rule D) is v1 default for empirical-roster replay. Top-K (rule C) and Bernoulli replay are **sensitivity** runs afterward.
Gibbs t: pick from reigning temperature sweep (`reigning_hero/calibration/temperature/`); start with $t = 1$ ($\log_{10} t = 0$) unless a bin tag says otherwise. Not the same as MLE $t^* \approx 1.07$.

SELECT history — what did we use before?

Short answer: For **generative sim** (assign → score → select → HERO), **yes — almost always deterministic top-K (rule C).** Gibbs (rule D) arrived with **PD20** as a deliberate upgrade/sensitivity test. **PD21 fitting** never used SELECT at all.

Four winner rules in code (`tier1_pool_assignment.py`)

Rule	Name	Mechanism	When we used it
C	Top-K	Take the K highest scores S_i	**Default everywhere** — `tier1_sim_config.py` `WINNER_SELECTION = "C"`; Pass A/B; 540 notebook; PD17 λ sweep; most grandchild diagnostics
D	Gibbs K-draw	$w_i \propto \exp(S_i/t)$, draw K without replacement	**PD20+** — temperature sweep, reigning calibration; **new v1 default** for empirical-roster replay
A	Proportional sample	K draws $\propto$ positive weights (not Gibbs)	Older 537 experiments; not main path
B	Bernoulli-ish	Independent coins, expected $\approx K$	Older 537 experiments; not main path

Cold Gibbs nests top-K: at very small t_{Gibbs}, rule D collapses to rule C (see `grandchild_temperature_cold_limit_diagnostic.py`).

Timeline (MBB modeling)

Phase	SELECT	Notes
Tier1 / 540 / Pass A–B	Top-K (C)	"Score then pick the K best" — BINDING v1 winner rule
PD17 (`grandchild_lambda_select_sweep`)	Top-K (C)	λ sweep on sim bridge; empirical roster caps
PD20 (`grandchild_temperature_select_sweep`)	Gibbs (D)	**Gate:** does inverted-U survive soft/stochastic select? **Yes** → proceed to MLE
PD21 (`pd21_draft_bernoulli_mle`)	**No SELECT**	Fit uses softmax → Bernoulli on real Y_i; top-K overlap is diagnostic only
Reigning calibration	Gibbs sweep (D) at $\rho = 0$	Same PD20 machinery on 09–21 panel
Empirical-roster replay (next)	**Gibbs (D)** — Charles lock	Then sensitivity: top-K, Bernoulli replay

So you were not wrong to think "top-K everywhere" — that **was** the generative default for a long time. Gibbs is the intentional next step for replay, building on PD20.

Open questions (remaining)

1. SELECT rule for sim HERO — decided

Fitting: Bernoulli + softmax (PD21) — unchanged.

Generative replay (v1): Gibbs K-draw (rule D) with K_s = empirical draft count per season.

Sensitivity (after v1): deterministic top-K (rule C); Bernoulli draws from MLE p_i.

Open sub-choice: which t_{Gibbs} from the reigning sweep (default start: $t = 1$).

2. What is L^C in the fit vs in the HERO readout?

- PD21 MLE on fixed rosters uses **`pool_c_smooth_team`** (smoothed team pool context), not necessarily raw `poolq_loo`.
- Reigning porch HERO uses **`poolq_loo`** on **PPM z** (leave-one-out teammate mean).

Inconsistency risk: fit on one local-context definition, plot on another.

Fix options: (a) refit MLE with LOO-based L^C for HERO alignment; or (b) build sim HERO using the **same** L^C column the MLE used. Do not mix silently.

3. HERO aperture must match exactly

Empirical reigning lock (slide 12):

- Seasons **2009–2021**
- **last-ps** rows for HERO bins
- **ever-Y** draft label
- **EW16** equal-width bins on `poolq_loo`
- **ALLT** (no +DFT filter on panel)
- min20 · mg10 · winsor 0.01–0.99 · PPM z

Sim HERO must copy this binning and row filter. Change one filter → you are comparing different objects.

4. Does γ^* in the generator match the MLE file?

γ sharpens viability / L^C in the full grandchild story. Confirm the draft generator reads **reigning** `gamma_hat` from the JSON above — not a stale default in `tier1_sim_config.py`.

5. What this path cannot claim

With frozen rosters you **cannot** claim:

- homophily / ρ calibration on MBB,
- "league formation" predictions,
- ASSIGN-layer Pass C ρ ablations.

You **can** claim:

- *Given real MBB rosters, do calibrated score + select reproduce draft-vs-LOO geometry?*

That matches parking homophily.

6. Alternate performance metrics (SCOUT ladder)

Kill criterion (Charles, Aug 28): A metric is only worth switching to if it **breaks** the naive **monotone increasing** draft-rate vs LOO plot. Higher H_{sort} alone (BPM, TS%) is **not** enough — it may even make $\hat{A}_i$ vs LOO **more** aligned within teams.

For this replay: **keep PPM** unless SCOUT shows a non-monotone HERO on another metric.

Suggested protocol (v1 — for implementation)

1. **Load** 2009–2021 empirical panel (all-ps for fitting context; last-ps for HERO comparison).
2. **Lock** $\rho = 0$ (documented; no ASSIGN sim).
3. **Lock** γ^*, λ^*, MLE t^* from reigning MLE JSON (not 2013–21 campaign defaults).
4. **SELECT (v1):** Gibbs rule D, K_s = empirical count, $S_i = A_i - \lambda^* L_i^C$, $w_i \propto \exp(S_i / t_{\text{Gibbs}})$; sensitivity runs: top-$K$, Bernoulli replay.
5. **Align** L^C definition between fit and HERO axis.
6. **Simulate** Y_i^{sim} on unchanged team assignments.
7. **Build sim HERO** — same EW16 LOO bins as empirical reigning hero.
8. **Side-by-side** emp vs sim: draft rate vs LOO (+ optional LPM β_2 for a scalar curvature tag).

Deliverable name (proposed): `reigning_hero_sim_hero/` under sandbox, script `reigning_hero_sim_hero.py` — empirical panel in, sim draft flags out, HERO PNG next to empirical.

Numbers to keep on one card

Quantity	Reigning 09–21	Notes
ρ^*	**0.0**	Bracket floor
H_{sort} (PPM, pooled)	**~0.064**	Weak team sorting
γ^*	**~19.57**	MLE
λ^*	**~1.30**	MLE (campaign was ~2.57 on 2013–21)
t^* (MLE / softmax)	**~1.07**	Fitted in PD21; sharpens A_i in logits
Gibbs SELECT t	**1.0 (v1 lock)**	Mid-t sweep: flat band; cold t excluded from default overlay
Sim HERO LPM β_2 $(t = 1)$	**≈ +0.002**	Matches emp flatness
Sim ever-drafted (last-ps)	**617**	Emp **615**
Empirical HERO LPM β_2	**~+0.0017**	Flat / not concave on PPM last-ps

Aug 28 end-of-day — what we learned (sim replay v1)

Tone check: No fireworks. Steady reality — which is what this test was for.

Headline

Given frozen 09–21 rosters and reigning MLE score knobs, Gibbs SELECT at $t_{\text{Gibbs}} \approx 1$ reproduces a *flat* empirical LOO-HERO at roughly the right draft count. That is a **pass on shape family** (flat vs inverted-U), not a proof that temperature cleanly tunes curvature.

Numbers (reigning lock)

Readout	Empirical	Sim (Gibbs $t = 1$, rule D)
Ever-drafted on last-ps	**615**	**617**
LPM β_2 on poolq$_{\text{LOO}}$	**+0.0017** (flat)	**≈ +0.002** (flat)
K_s source	NBA lookup `draft_year == season`	same (not `sum(Y)` on all-ps)

Artifacts: `reigning_hero/sim_hero/REIGNING_SIM_HERO_gibbs_t1_*` and overlay `REIGNING_SIM_HERO_gibbs_t_sweep_*_last_ps.png`.

Gibbs t sweep — what it did and did *not* show

- **Cold t** ($\lesssim 0.5$): $\beta_2 \approx +0.03$, left-heavy HERO — **top-K-like pathology** on score S, read out on LOO axis. Documented; excluded from default mid-t overlay.
- **Mid–hot t** (0.75–15): all sim curves stay in a **±0.02 band** around empirical flatness; **no monotone “temperature dial”** on β_2.
- **Interpretation:** reshuffling ~617 discrete picks + **SCORE axis ≠ HERO axis** (L^C from `pool_c_smooth_team` in S; HERO bins `poolq_loo`). Wiggling β_2 with t is mostly **weak identification / bin noise**, not structured physics — except at cold t.
- **v1 lock stands:** $t_{\text{Gibbs}} = 1$ (near MLE board temperature; best β_2 match in sweep). Not uniquely pinned by curvature alone.

Axis mismatch (open, not a code bug)

Layer	Object on LOO?	Notes
SCORE	No — team L^C at γ^*	`pool_c_smooth_team` in PD21 MLE
HERO	Yes — poolq_{LOO}	reigning slide 12 lock

BDP porch (Aug 28): $\hat{T}_j$ wide vs LOO tight; $H^{\text{team}}_{\text{sort}} \approx +0.06$ but $H^{\text{LOO}}_{\text{sort}}$ can go **negative** (`frac_var_loo_residual > 1`). Cold Gibbs over-drafts low-LOO "big fish" — **feature of mismatch**, not implementation error.

Next sensitivity (when ready): SELECT with `pool_c_smooth_loo` in S, same λ^*; or refit MLE on LOO L^C. See `reigning_hero/README.md`.

SELECT rule comparison (same replay)

Rule	Verdict
Gibbs D, $t = 1$	**v1 default** — flat HERO, right K
Top-K (C)	$\beta_2 \approx +0.03$ — same cold pathology as low t
Bernoulli replay	~13 sim picks — wrong generative story for fixed K

Alex one-liner (Aug 28 evening)

> *On real MBB rosters we parked ρ, fit score on the frozen panel, and simulated draft with Gibbs SELECT.* ***Empirical and sim HERO are both flat*** *at ~617 picks; $t_{\text{Gibbs}} \approx 1$ is a reasonable lock. Temperature does* ***not*** *smoothly tune LOO curvature — score uses team congestion, HERO reads LOO. Cold select looks like top-K and breaks LOO geometry; that's the main structured failure mode.*

What we did *not* get today

- A crisp "sweep t → find inverted-U" story on empirical replay (empirical isn't inverted-U).
- Unique t^* from β_2 alone.
- Resolution of team vs LOO in L^C without a deliberate sensitivity run.

What we *did* get

- **Generative replay pipeline** (`reigning_hero_sim_hero.py`) with sweep overlay, career cap, correct K.
- **Honest kill on cold-t / top-K as default.**
- **BDP deck** extended (H_{sort}, residuals, $\hat{T}_j$ vs LOO, no-winsor overlay) — explains *why* axis mismatch matters before the sim slide.

SCOUT / perf-metric work (parallel, optional)

Disposable folder: `population_sandbox/_DISPOSABLE_perf_metric_rho_eda/`

- **H_sort ladder** asks: does another metric show more team sorting?
- **Charles gate:** only promote a metric if **draft vs LOO** is not naive monotone.
- Does **not** block the empirical-roster replay on PPM.

Thread log

Date	Entry
2026-08-28	Alex: park homophily on MBB; fit λ, γ, t on empirical teams; generate draftees; compare HERO. Document written for print/PDF.
2026-08-28	Expanded "softmax vs Bernoulli vs Gibbs" deep dive — clarify two temperatures and what is already decided.
2026-08-28	Added formula sheet (η, softmax, Bernoulli ℓ, Gibbs weights, HERO bins).
2026-08-28	Added plain-language "board logits" glossary (after formula sheet).
2026-08-28	Charles lock: Gibbs K-draw (rule D) v1 for replay; SELECT history section.
2026-08-28	Implemented `reigning_hero_sim_hero.py` — Gibbs/topk/bernoulli replay in `sim_hero/`.
2026-08-28 (eve)	**Sim replay v1 result:** flat emp + sim HERO; 615 vs 617 picks; $t_{\text{Gibbs}} = 1$ lock; β_2–t oscillation = weak ID + axis mismatch. Section added above.
2026-08-28 (eve)	Gibbs t overlay PNG; cold t pathology; BDP porch slides (H_{sort}, LOO residuals, $\hat{T}_j$ vs LOO no-winsor).

PDF conversion (Charles)

From repo root:

```
./scripts/convert_single_md_to_pdf.sh \
  "3-Master_Plan/re_entry/HEROs_and_PASSes/MBB_empirical_roster_select_replay.md"
```

Optional narrow style if you use it elsewhere: add `pdf_styles_narrow.css` per `3-Master_Plan/re_entry/MARKDOWN_FOR_PDF_PLAYWRIGHT.md`.